

LAB 10 – Authentication, Session Management & Notifications

1. Overview

This lab guides students through building authentication and user management features in Flutter step by step.

Each lab task is implemented as a **separate runnable Flutter project**, allowing students to verify each feature independently.

After completing all parts, students integrate all features into a final full project.

Local Notification is a **mandatory part** of this lab to ensure students achieve **LO7 – integrating notifications in mobile applications**.

2. Learning Outcomes

After completing this lab, students will be able to:

- Implement authentication flows using mock backend and real REST APIs
- Manage user sessions using local persistence
- Implement auto-login and logout mechanisms
- Integrate Firebase Authentication (Google Sign-In)
- Integrate local notifications after key user actions (LO7)

3. Prerequisites & Tools

- Android Studio
- Android Emulator or physical device (API 33+ recommended)
- Flutter SDK (stable channel)
- Internet connection (required for REST API & Firebase)

4. Required Packages & References

- DummyJSON Authentication API
<https://dummyjson.com/docs/auth>
- SharedPreferences
https://pub.dev/packages/shared_preferences
- Firebase Authentication
<https://firebase.google.com/docs/auth>
- Firebase packages
firebase_core, firebase_auth, google_sign_in
- Flutter Local Notifications
https://pub.dev/packages/flutter_local_notifications

5. Submission Rules

5.1 Required Submissions

Students must submit **ALL** of the following:

Part projects (each runs independently):

- Lab10_1_MockLogin
- Lab10_2_RealApiLogin
- Lab10_3_AutoLogin_Logout
- Lab10_4_FirebaseGoogleSignIn
- Lab10_5_Notification

Final integrated project:

- Lab10_Full

5.2 Naming Convention

- Lab10_1_MockLogin
- Lab10_2_RealApiLogin
- Lab10_3_AutoLogin_Logout

- Lab10_4_FirebaseGoogleSignIn
- Lab10_5_Notification
- Lab10_Full

5.3 Submission Format

- Each project must be compressed as a .zip file
- Each submission should include a short README describing:
 - How to run the app
 - Test account (if any)

6. LAB TASKS

Lab 10.1 – Mock Login (Backend Simulation)

Goal

Build a login screen with validation and simulate backend authentication.

Description

Students create a login UI and simulate a backend authentication process using a mock service. This lab focuses on UI structure, validation, and basic authentication flow.

Steps

1. Create project Lab10_1_MockLogin
2. Build Login UI (email, password, login button)
3. Validate input using Flutter Form
4. Simulate authentication with a delay and mock token
5. Navigate to Home screen on success

Expected Output

- Valid credentials navigate to Home

- Invalid credentials display an error message

Lab 10.2 – Real REST API Login

Goal

Authenticate users using a real REST API (DummyJSON).

Description

This lab replaces mock authentication with a real HTTP login request. Students practice handling API responses, loading states, and errors.

Steps

1. Create project Lab10_2_RealApiLogin
2. Send POST login request to DummyJSON
3. Parse token from response
4. Handle loading and error states
5. Navigate to Home on success

Expected Output

- Login succeeds using real API
- Errors are handled gracefully

Lab 10.3 – Auto Login & Logout

Goal

Persist authentication state and restore login automatically.

Description

Students implement session persistence using SharedPreferences and manage login state across app restarts.

Steps

1. Create project Lab10_3_AutoLogin_Logout

2. Save token after login
3. Implement SplashScreen routing
4. Auto-login if token exists
5. Logout clears session and returns to Login

Expected Output

- App skips login when token exists
 - Logout forces re-authentication
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Lab 10.4 – Firebase Authentication (Google Sign-In)

Goal

Integrate Google Sign-In using Firebase Authentication.

Description

Students configure Firebase and implement Google Sign-In as an alternative authentication method.

Steps

1. Create project Lab10_4_FirebaseGoogleSignIn
2. Configure Firebase project and SHA keys
3. Implement Google Sign-In flow
4. Display user profile information
5. Logout from Firebase and Google

Expected Output

- Google Sign-In works correctly
 - User profile is displayed
 - Logout returns to Sign-In screen
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Lab 10.5 – Local Notification (MANDATORY – LO7)

Goal

Integrate local notifications in a Flutter application.

Description

Students integrate local notifications to enhance user experience after important user actions.

Steps

1. Create project Lab10_5_Notification
2. Setup flutter_local_notifications
3. Request notification permission (Android 13+)
4. Trigger notification manually for demonstration
5. Verify notification is displayed by the system

In the integrated project, the same notification method must be called after a successful user action (e.g. login success).

Expected Output

- Notification appears on the device
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Lab10_Full – Final Integrated Project

Goal

Integrate all authentication and notification features into one complete application.

Integration Order (Recommended)

1. Base project from Lab 10.1
2. Replace mock login with real API login (Lab 10.2)
3. Add session persistence and Splash routing (Lab 10.3)
4. Add Firebase Google Sign-In (Lab 10.4)
5. Add Local Notification (Lab 10.5)

Minimum Required Features

- SplashScreen routing

- Real API login
 - Auto-login and logout
 - Firebase Google Sign-In
 - Local notification integration
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7. Final Checklist

- All 5 part projects run independently
 - Lab10_Full integrates all required features
 - Local notification works correctly
 - Correct naming and submission format
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8. Score Distribution (Percentage)

- Lab 10.1 – Mock Login: 15%
- Lab 10.2 – Real REST API Login: 20%
- Lab 10.3 – Auto Login & Logout: 25%
- Lab 10.4 – Firebase Google Sign-In: 20%
- Lab 10.5 – Local Notification (LO7): 10%
- Lab10_Full – Integrated Project: 10%

Total: 100%